

RANA PUBLIC SCHOOL

CLASS VIII – MATHEMATICS WORKSHEET

Half-Yearly Revision Worksheet | Objective & Subjective Questions

Instructions: Answer all questions. No working space is provided on this worksheet. Show necessary steps in your notebook.

SECTION A – OBJECTIVE TYPE QUESTIONS

A. Choose the correct option. (1–15)

1. Which of the following is equal to $(\frac{3}{5})^3$? (a) $\frac{27}{125}$ (b) $\frac{9}{25}$ (c) $\frac{15}{8}$ (d) $\frac{125}{27}$
2. Which is the square root of 289? (a) 14 (b) 16 (c) 17 (d) 19
3. Which of the following is NOT a perfect cube? (a) 512 (b) 4096 (c) 216 (d) 367
4. Which of the following is divisible by 9? (a) 725 (b) 738 (c) 742 (d) 751
5. Simplify: $18x^2y^2z^2 \div 6xyz$. (a) $3xyz$ (b) $3xy$ (c) $12xyz$ (d) $3x^2y^2$
6. The HCF of $12x^2y$ and $18xy^2$ is: (a) $3xy$ (b) $6xy$ (c) $6x^2y^2$ (d) $12xy$
7. Which equation has $x = 8$ as its solution? (a) $x + 5 = 3$ (b) $x - 3 = -5$ (c) $\frac{x}{4} = -2$ (d) $x - 5 = 3$
8. How many edges does a cube have? (a) 6 (b) 8 (c) 10 (d) 12
9. The degree of the polynomial $7x^3 - 2x + 5$ is: (a) 1 (b) 2 (c) 3 (d) 5
10. Which identity is correct? (a) $(a+b)^2 = a^2+b^2$ (b) $(a+b)^2 = a^2+2ab+b^2$ (c) $(a+b)^2 = a^2-2ab+b^2$ (d) $(a+b)^2 = a^2-b^2$
11. Which of the following is a monomial? (a) $x+2$ (b) x^2+y^2 (c) $7xyz$ (d) x^2+x+1
12. The reciprocal of $-\frac{5}{7}$ is: (a) $\frac{5}{7}$ (b) $-\frac{7}{5}$ (c) $\frac{7}{5}$ (d) $-\frac{5}{7}$
13. Which sequence follows the rule 'multiply by 3'? (a) 2, 5, 8, 11 (b) 3, 9, 27, 81 (c) 10, 20, 30, 40 (d) 25, 20, 15, 10
14. Euler's formula for a polyhedron is: (a) $V+F-E=2$ (b) $V+E-F=2$ (c) $V+F+E=2$ (d) $E-V-F=2$
15. Which number is in standard form? (a) 0.0000042 (b) 42×10^{-7} (c) 4.2×10^{-6} (d) 420×10^{-8}

B. State whether the following statements are True or False.

16. $8ab$ and $-10ba$ are like terms.
17. $5(x+1)$ is a polynomial.
18. The square of a number means raising it to the power 2.
19. Zero can be multiplied by any rational number.
20. A linear equation in one variable has only one unknown.

21. The cube of an even number is even.
22. The highest power of a variable in a polynomial is called its degree.
23. A ball is a three-dimensional object.
24. An equation is a mathematical statement showing equality between two expressions.
25. The number of vertices of a triangular prism is 6.

C. Fill in the blanks.

26. The next term in 30, 25, 20, 15, ____ is ____.
27. The next term in 3, 9, 27, 81, ____ is ____.
28. $\sqrt{196} = \underline{\hspace{1cm}}$.
29. $\sqrt[3]{1331} = \underline{\hspace{1cm}}$.
30. Any non-zero number raised to the power 0 is ____.
31. The reciprocal of $\frac{2}{9}$ is ____.
32. A cuboid has ____ faces.
33. A cube has ____ vertices.
34. The number of edges in a tetrahedron is ____.
35. In a polynomial, the term with the greatest exponent has the highest ____.

SECTION B – VERY SHORT ANSWER QUESTIONS

1. Find the cube root of 512.
2. Find the possible digit a if $32a5$ is divisible by 3.
3. Divide: $57xy \div 3z$.
4. Find the HCF of $21x^2y^2z$ and $7xyz$.
5. Find x if $2x/0.032 = -100$.
6. Write the number of faces, edges and vertices of a cube.
7. Write the number of faces, edges and vertices of a tetrahedron.
8. Express 0.00000067 in standard form.
9. Express 12,54,36,000 in standard form.
10. Write 8.99×10^{-7} in usual form.
11. Find the Pythagorean triplet whose one member is 7.
12. How many integers lie between 9^2 and 10^2 ?

13. Find the number of edges of a polyhedron having 10 faces and 12 vertices.
14. Find the magic sum of a 3×3 magic square whose middle number is 50.
15. Complete the identity: $a^2 - b^2 = \underline{\hspace{2cm}}$.

SECTION C – SHORT ANSWER QUESTIONS

1. Solve: $2(0.25y - 1) + 1.5y = 8$.
2. Solve: $7x + 3 = 2x + 13$.
3. Find the number which is 20 more than one-fifth of it and equals 120.
4. Factorise: $x^2 + 12x + 36$.
5. Factorise: $x^2 - x - 2$.
6. Factorise: $36z^2 - 49x^2$.
7. Expand using an identity: $(2x + 3)^2$.
8. Find the product: $(3x^2y)(-2xy^3)$.
9. Find the product: $9xy(x^2y - xy^2)$.
10. Add: $(m^3 + 3m^2 - 18m)$, $(23m - 4m^2)$, and $(5m^3 - 1)$.
11. Find the product: $(x^3y)(x^2y^2)$.
12. Find the area of a rectangle whose length is $(2x+7)$ units and breadth is $(x-5)$ units.
13. Simplify: $(a+b)(a-b)$.
14. Verify $(a+b)^2 = a^2+2ab+b^2$ for $a=3$ and $b=4$.
15. Using an identity, simplify $109^2 + 91^2$.

SECTION D – LONG ANSWER / APPLICATION QUESTIONS

1. A perimeter of an isosceles triangle is 252 cm. If its unequal side is 72 cm, find the length of each equal side.
2. What should be added to $a^2 + 7a$ to obtain $9a^2 - 14a$?
3. Find the number of integers between 39^2 and 40^2 .
4. Find the length of each side of a cube whose volume is $8,04,357 \text{ cm}^3$.
5. Construct a 4×4 magic square starting from 20 and find its magic sum.
6. Classify each polynomial as monomial, binomial or trinomial: (i) $x^2+y^2+z^2$ (ii) 100 (iii) xyz^2+x^2yz (iv) $a^2+b^2+c^2$ (v) $p+2$.
7. Divide: $(10x^2+15x+5) \div (10x+5)$.
8. Divide: $(a^2+2ab+b^2) \div (a+b)$.

9. Simplify using identities: $(200)^2 - (100)^2$.
10. A polyhedron has 10 faces and 12 vertices. Use Euler's formula to find its number of edges and verify the result.
11. Draw a pentagonal prism and state its number of faces, edges and vertices.
12. Draw the top, front and side views of a cuboid and a cylinder.

SECTION E – MIXED REVISION

1. Write the next two terms: 80, 65, 50, 35, ____, ____.
2. Write the next two terms: 20, 40, 80, 160, ____, ____.
3. Find the value of a in $3 \times 5^2 \times a = a \times 3 \times 25$.
4. Find the missing number: $555 \times 6 = 3330$.
5. Find the value of a if $a \times a \times a = a$ and a is a whole number.
6. Write the first five terms of the Fibonacci sequence beginning 1, 1.
7. State the number of faces of a triangular prism.
8. State the number of vertices of a pentagonal prism.
9. Find the HCF of $8xy$ and $16x^2$.
10. Find the quotient: $12x^2y^2z^2 \div 6xyz$.
11. Express 354 in standard form.
12. Find a Pythagorean triplet containing 6.
13. Factorise: $b^2m - b^2n$.
14. Complete: $(a-b)^2 = \underline{\hspace{2cm}}$.
15. Explain why 723 is divisible by 3 but not by 2.